MIT SCHOOL OF COMPUTING
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A Leap Towards World Class Education

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Subject: IDS Roll No.: 28 Exp. No.: 1

Name of the Experiment:

Performed on:

Submitted on:

Marks	Teacher's Signature with date
10	

1] Define an algorithm. List any 4 Characteristics of an algorithm.

→ An algorithm is a finite sequence of well defined. Step-by-step instruction used to solve a specific problem or perform a computation.

Characteristics:

1. Input: An algorithm takes zero or more inputs
2. Output: It produces at least one output
3. Definiteness: Each step is clear, precise, unambiguous
4. Finiteness: The algorithm terminates after a finite number of steps.

2] Define data structure and Abstract Data type (ADT). give one example of each

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

→ A data Structure is a way of organizing Storing and managing data in memory that it can be accessed and modified efficiently.

An Abstract data types (ADT) is logical description of a data type that specifies the operations that can be performed on it without specifying how these operations are implemented.

Example:-
- Data Structure: Array
- ADT: Stack (operations: Push, POP, Peek)

3.] Explain the process of converting a problem into a program. Illustrate the role of algorithms in this process.

→ 1.] Problem defination: Understand requirements and constraints.
2. Analysis: Identify inputs, outputs and processing
3. Algorithm design: create step-by-step logic to solve the problem.
4. FlowChart / Pseudocode: represent the algorithm visually or textually.
5. Coding: Convert algorithm into a program using a programming language.

6. Testing & Debugging : Check Correlatio and remove errors.
7. Documentation & maintenance : Impr ve & maintain the program.

Role of algorithm :
The algorithm acts as a blueprint of the solution. It ensures correct ness, efficiency and makes coding easier and error free.

4. Explain linear and Non-linear data structures with suitable examples
→ Linear data structure :
Data elements are arranged sequenti
- Army
- Stack
- Quene
- Linked list

Non-linear data structure :
Data elements are arranged in a hierdrchical or interconnected mann er

Ex → Tree
Graph.

5.] Differentiate b/w algorithm and program analyze their roles in SDLC

→

Algorithm	Program
Step-by-Step logic	Written code
language independent	language dependent
Design level	Implementation level
No syntax rules	follow syntax rule

Role in SDLC:

- Algorithm: used in design phase to plan the Solution.
- program: used in implementation phase to execute Solution on Computer.

6.] Examine the relationship among data, data structure and algorithm. Justify how the choice of data Structure affects algorithm efficiency

→ Relationship:

- Data is raw information
- Data Structure organizes the data
- Algorithm processes the data using data structures.

Justification:

The efficiency of an algorithm

depends on the data structure u

- Searching is an array → slower (
- Searching in a binary search h faster (O(logn))

Conclusion:
choosing the right data structure improves time and space efficiency of Algorithms.